

GRIDLOCK 2.0

AI Traffic Command Center: Predict • Plan • Prevent

Engineered by
The Query Squad

• Trained on 8,173 historical
Bengaluru Astram traffic events

• Targeting Theme 2:
Event-Driven Congestion

The Core Challenge: Event-Driven Congestion

Urban traffic incidents rapidly cascade into corridor-wide gridlock. Current systems lack the analytical foresight to prevent this chain reaction.

Planned Events (5.7%)

Political rallies, VVIP movements, stadium crowds, and scheduled infrastructure maintenance.

Unplanned Events (94.3%)

Sudden vehicle breakdowns, waterlogging, tree falls, and accidents.

Impact of the Status Quo: Delayed officer deployment, increased congestion duration, avoidable CO₂ emissions, and hindered emergency responses.



Shifting from Blind Reaction to Transparent Command

Event Impact

Not quantified in advance; assumed based on historical memory.

Mathematically forecasted (Score **1-100**) before escalation.

Resource Deployment

Experience-driven guesswork; reactive dispatch.

Data-driven allocation of specific manpower, barricades, and tow units.

System Learning

No formal post-event feedback loop; mistakes repeat.

Continuous **ML** feedback loop calibrated on **8,173** operational records.

Current Reality

The Future State (Gridlock 2.0)

Grounded in Reality: The Astram Dataset

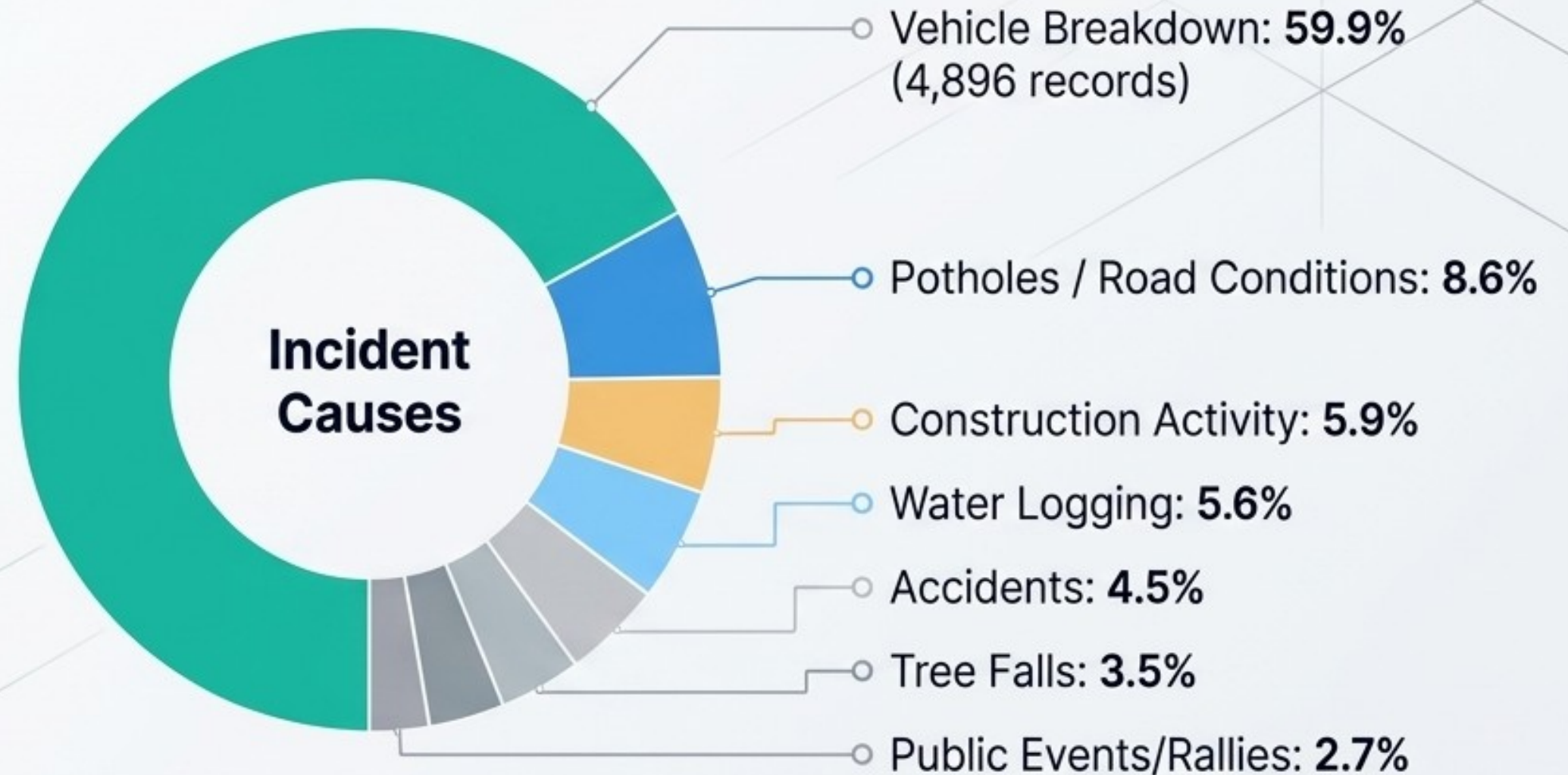
Total Foundation

8,173 traffic incidents directly from the Bengaluru Traffic Police Astram System.

Dimensionality

46 raw fields covering spatial (lat/lon, zones), temporal (datetime), and categorical descriptors.

High-density arterial corridors like Bellary Road (989 events) and Mysore Road (743 events) represent extreme-vulnerability nodes compared to the general city grid.



Engineering Integrity: The 'Honest R²' Approach

Inflated hackathon metrics fail in production. We explicitly removed target leakage to ensure the model only uses pre-event signals.



Pre-Event (Valid Inputs):

Location, Time,
Priority, Cause,
Vehicle Type.

Post-Event (Excluded Vectors):

duration_minutes: Unknown at the time
of forecast.
status (open/closed): A post-event
operational outcome.

Takeaway: An R² of 0.253 from only pre-event signals is realistic, highly predictive for traffic, and operationally trustworthy.

The Integrity Metric

Inflated R²:

0.825

(With future variables)

Honest R²:

0.253

(Pre-event signals only)

The 5-Layer Integrated Intelligence Architecture



Data Ingestion (`data_loader.py`): Normalizes Astram CSV, parses datetimes, maps corridors, and extracts zones.

Feature Engineering (`features.py`): Derives 20+ ML-ready inputs; computes haversine segment lengths; encodes temporal signals.

Impact Forecasting (`train.py` + `predict.py`): 5-fold CV Random Forest ensemble generating an impact score (1–100).

Resource Allocation (`resource_model.py`): Multi-output RF pipeline outputting specific manpower, barricade, and diversion targets.

XAI Copilot + Telemetry (`xai.py`, `telemetry.py`): Natural language prediction rationale and physics-based environmental mathematics.

Architecture Principle: Zero inference-time components use information unavailable before an event begins.

Capturing Pre-Event Signals: Feature Engineering



Categorical Drivers

Event Type & Cause: Differentiates clearance profiles (planned staging vs. reactive surge for breakdowns).

Corridor Identification: 14 named arterials vs. non-corridor grid, dictating free-flow capacity.

Priority (High/Low): Applies a baseline 1.4× multiplier in severity.

Vehicle Type: Heavy vehicles exponentially increase blockage probability.

Numerical & Derived Drivers



Temporal: **start_hour**, **is_peak_hour** (binary trigger for 8-9am / 5-7pm), **is_weekend**, **start_month** (monsoon waterlogging).



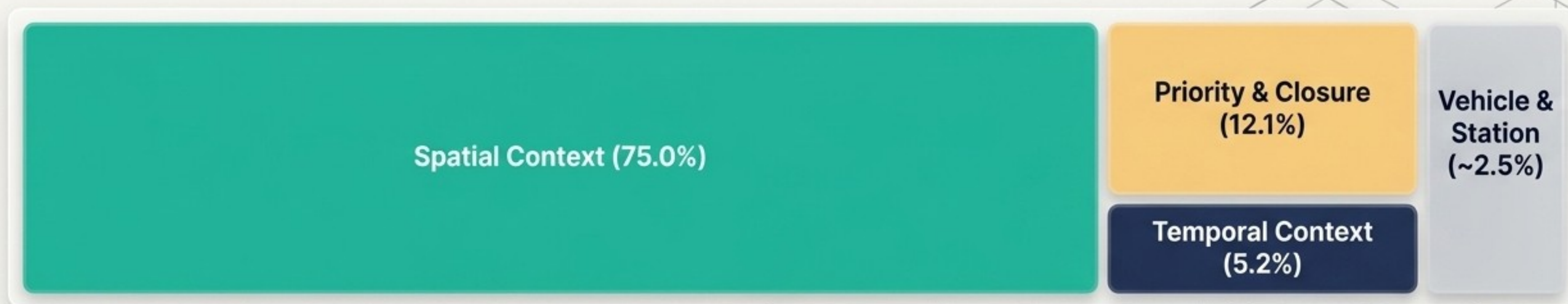
Spatial Physics: **segment_length_km** computed via Haversine formula mapping start/end GPS coordinates to gauge physical lane blockage.



Operational: **requires_road_closure** (carries a +0.35× multiplier, the strongest single operational driver).

Feature Importance: How the AI Weighs Reality

(Mean Decrease in Impurity across 250 trees)



Spatial Context (75.0%):

Location dictates impact. Raw GPS coordinates (latitude: 27.41%, longitude: 13.78%) and corridor mapping (non-corridor: 33.84%) confirm the model natively learned Bengaluru's street grid.

Temporal Context (5.2%):

Day-of-week and peak-hour data act as modifiers on top of spatial severity.

Priority & Closure (12.1%):

Operational severity flags heavily dictate resource dispatch. (priority_high: 9.09%, requires_road_closure: 2.98%).

Vehicle & Station (~2.5%):

Drives clearance complexity and officer proximity.

Key Insight: The dominance of raw latitude/longitude proves the system learned corridor-level flow patterns without requiring explicit human-labeled metadata.

Infrastructure of Powering the Prototype

Data Source



Astram Dataset

The foundation of the system, comprising 8,173 anonymized historical traffic events from the Bengaluru Traffic Police.

Backend & ML Pipeline

Core Web Framework



Powered by Python 3.11 and Flask 3.0 to manage high-concurrency requests and model serving.

Data Science Stack



Utilizes scikit-learn and LightGBM for predictive modeling, with pandas and numpy for high-speed data manipulation.

Model Integrity



Implements advanced feature engineering and leakage-free training protocols to ensure forecasting honesty and accuracy.

Impact Score Prediction



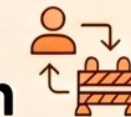
Module that calculates a 0–100 congestion severity score based on input parameters.

XAI Reasoning Engine



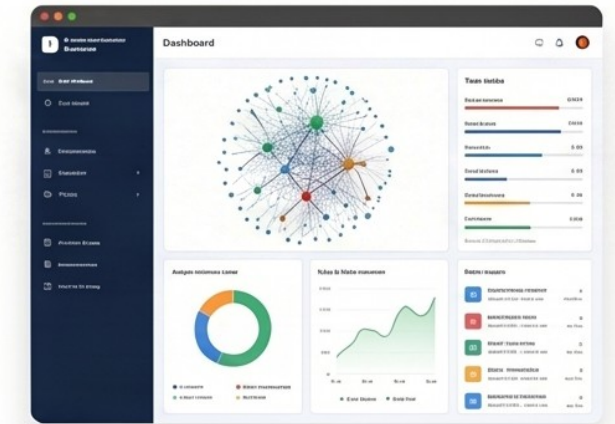
An Explainable AI component that provides plain-language justifications for every system recommendation.

Resource Recommendation



Automatically determines the necessary number of officers, barricades, and tow units for an event.

Frontend UI



AI Command Center Dashboard

HTML5/CSS3/JS, Jinja2, Maps API powering the Dynamic Telemetry UI.

Dynamic Telemetry UI

Integrates the Google Maps API to provide live spatial awareness and real-time alternate route diversion planning.

The Modeling Pipeline: Dual-Engine Intelligence

Engine 1: Impact Forecast Model

Algorithm: Random Forest Regressor (sklearn) / Auto-upgrade to LightGBM.

Configuration: 250 estimators, max depth 12 (prevents overfitting on sparse junctions).

Why: Natively handles high-cardinality categoricals (50+ junctions) combined with numerical temporal data. Robust against outlier severity spikes.

Engine 2: Multi-Output Resource Model

Simultaneously predicts 4 operational targets without ad-hoc lookup tables:

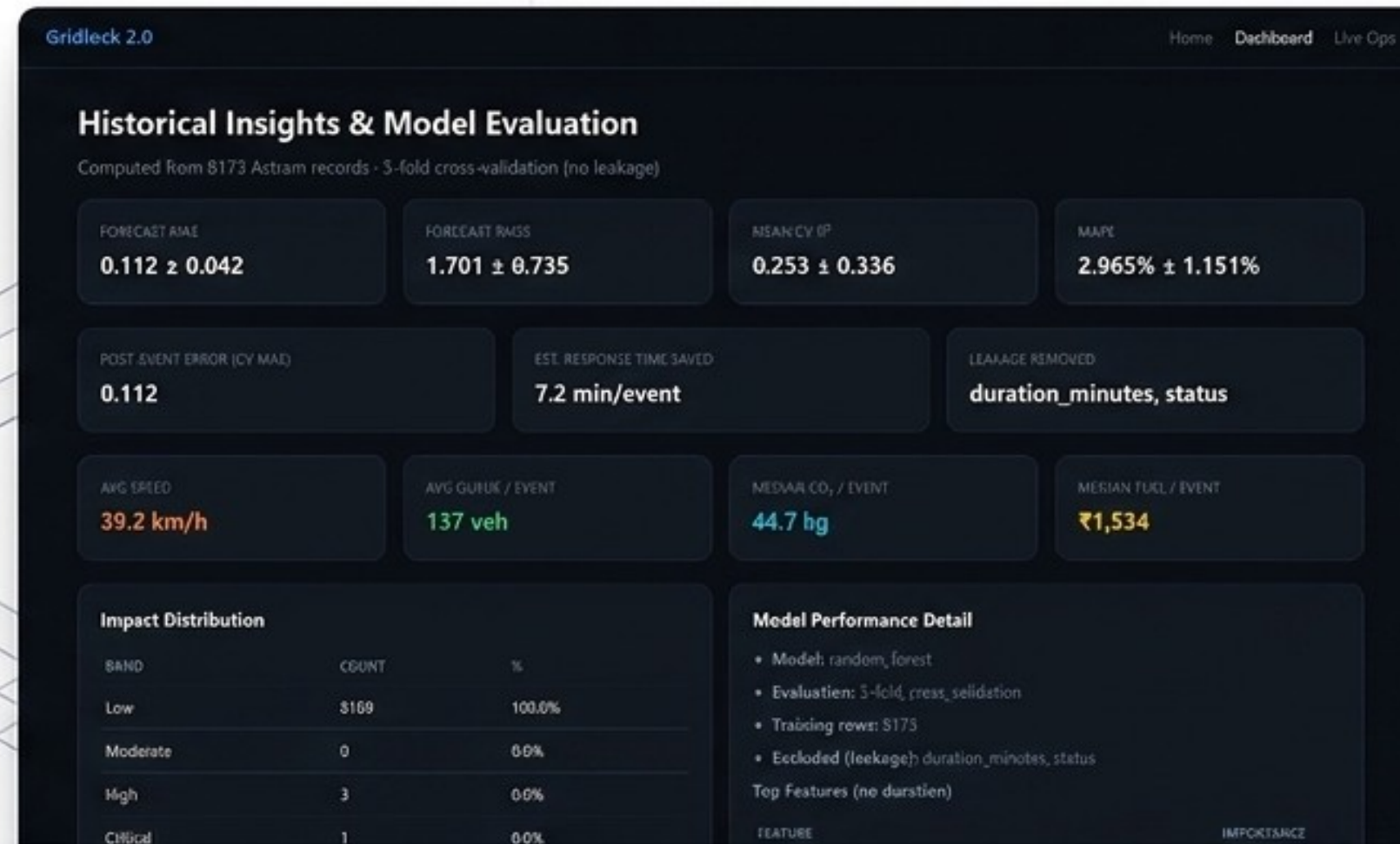
Manpower (Officers) ± 0.10 officers (RMSE)

Barricades ± 0.09 barricades (RMSE)

Clearance Time ± 1.09 minutes (RMSE)

Diversion Level 99.8% classification accuracy

Rigorous Evaluation: 5-Fold Cross Validation



5-Fold CV Strategy:

Guarantees metrics reflect generalizable real-world performance, not in-sample memorization.

Forecast MAE (0.112 ± 0.042):

Mean absolute prediction error on the 1–100 impact scale. High precision for severity scoring.

Forecast RMSE (1.701 ± 0.735):

Heavily penalizes massive predictive misses, ensuring officers are never dangerously under-deployed.

CV MAPE (~3%): Scale-invariant error measurement ensuring operational reliability for triage across both low-volume roads and high-density arterials.

The Physics Engine: Telemetry & Environmental Math

Gridlock 2.0 goes beyond ML scoring by mathematically converting predictions into concrete civic and environmental telemetry.

Speed Degradation

Average Speed =
Corridor Free-Flow
Speed × (1 - Impact
Reduction)

*Free-flow calibrated per-
corridor from the 60th
percentile of historical records.*

Queue Formation

Queue Length =
Traffic Flow Rate
× Delay ×
Congestion Factor

Emissions Reality

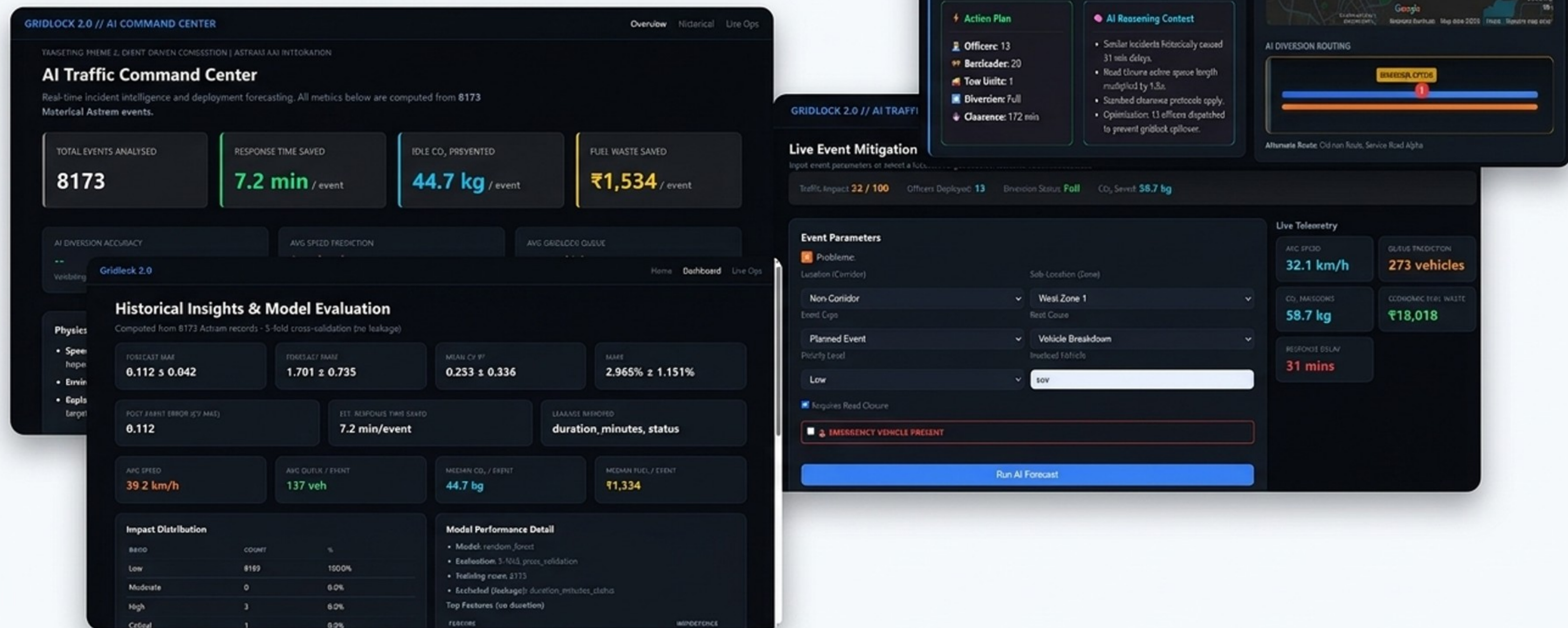
Idle CO₂ (kg) =
Queue_Vehicles ×
Idle_Minutes ×
0.0067

*Using IPCC standard idle
emission factor.*

Economic Burn

Fuel Cost (INR) =
(CO₂_kg ÷ 2.68 kg/L)
× ₹92/L

The Platform Modules: Real-Time Command Interface

**Traffic Impact** 22 / 100 **Officers Deployed** 13 **Diversion Status** Full **CO₂ Saved** 36.7 kg**Action Plan**

- Officers: 13
- Baricade: 20
- Tow Units: 1
- Diversion: Full
- Clearance: 172 min

AI Reasoning Context

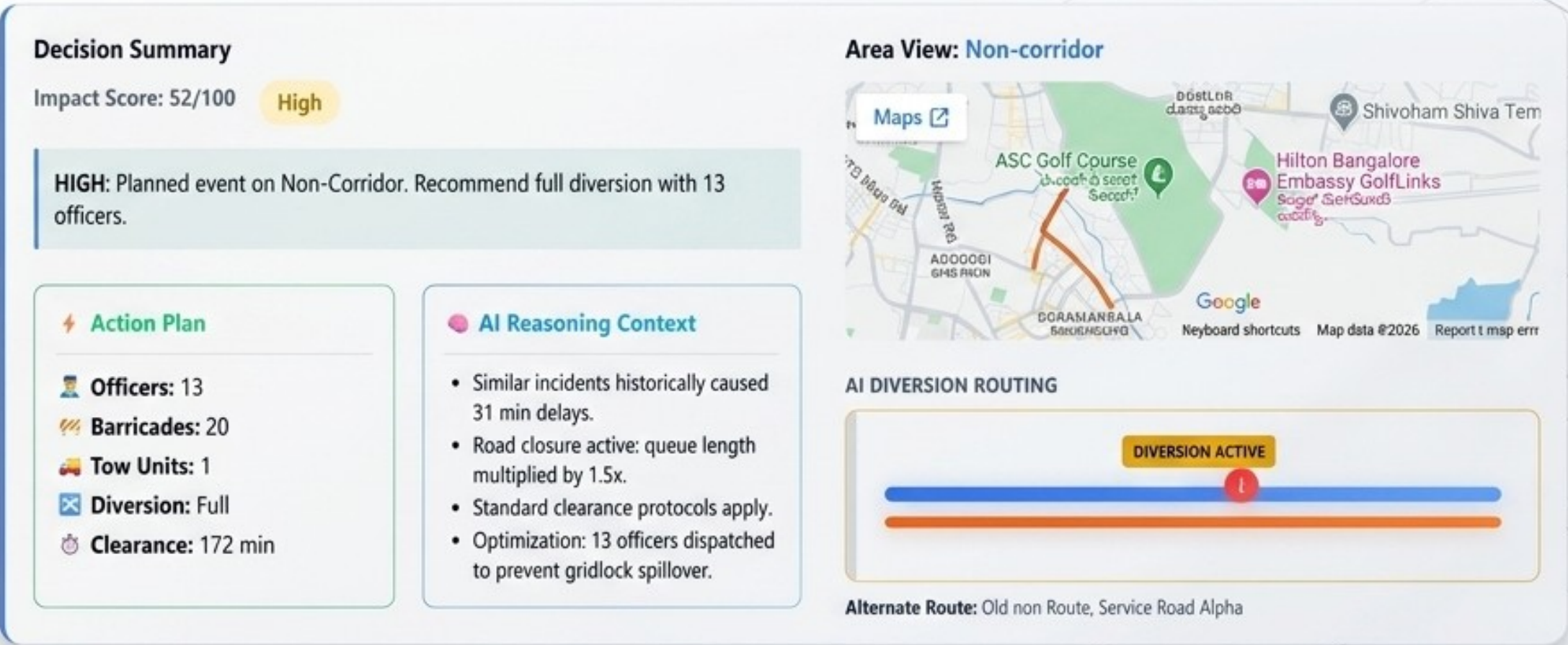
- Similar incidents historically caused 31 mins delays.
- Road closure active space length multiplied by 1.5x.
- Standard clearance protocols apply.
- Optimization: 13 officers dispatched to prevent gridlock spillover.

AVG SPEED
32.1 km/h**QUEUE PREDICTION**
273 vehicles**CO₂ MAINTENANCE**
58.7 kg**COGNITIVE TRAIL WASTE**
₹18,018**RESPONSE SLAY**
31 mins

Built with Python/Flask, scikit-learn, LightGBM, HTML5/CSS3/JS, Jinja2, Google Maps API.

Explainable AI: The Operator Copilot

Officers do not blindly follow a black-box score; they are given the complete rationale.



Operational Reality: The Live Ops Console

Parameter Inputs:

Dropdowns for precise spatial (Zone/Corridor) and operational (Cause/Priority) parameters.

Emergency Toggle:

A dedicated visual Red Line for immediate protocol overrides.

Instant Forecasting:

A single click generates an impact score, deployment count, and required barricades.

Live Telemetry Panel:

Instantly updates with average speed, queue prediction, CO₂ emissions, and economic fuel waste.

GRIDLOCK 2.0 // AI TRAFFIC COMMAND CENTER Overview Historical Live Ops SYSTEM ONLINE

Live Event Mitigation

Input event parameters or select a location to generate AI resource recommendations.

Traffic Impact: **52 / 100** Officers Deployed: **13** Diversion Status: **Full** CO₂ Saved: **58.7 kg**

Event Parameters

Location (Corridor)

Non-Corridor

Event Type

Planned Event

Priority Level

Low

☒ Requires Road Closure

☒ **EMERGENCY VEHICLE PRESENT**

Sub-Location (Zone)

West Zone 1

Root Cause

Vehicle Breakdown

Involved Vehicle

SUV

Run AI Forecast

Live Telemetry

AVG SPEED
32.1 km/h

CO₂ EMISSIONS
58.7 kg

RESPONSE DELAY
31 mins

QUEUE PREDICTION
273 vehicles

ECONOMIC FUEL WASTE
₹18,018

The "Red Line" Emergency Override

A dedicated code path for life-safety scenarios. When an ambulance, fire service, or police response vehicle is present, standard optimization is bypassed.

Standard AI Loop

Input Parameters



Impact Score Forecast



Optimized Resource Math



Standard Deployment

Emergency Priority Mode (Override)

Input: Emergency Vehicle Toggle Activated



Override: Impact Score forced to 99 (Critical)



Action: Triggers Priority Green Corridor



Deployment: Maximum Manpower (12–18 officers), Maximum Barricades (30–50). Immediate alternate route clearance commanded.

Engineered for the Real World



Leakage-Free Integrity

Identified and corrected target leakage mid-development, choosing scientific honesty over inflated optics.



Real Data Foundation

Grounded entirely in true Bengaluru Traffic Police operational records, learning **genuine** local corridor behaviors.



Multi-Output Resource Pipeline

Eliminates static look-up tables; every officer, barricade, and diversion strategy is **dynamically** derived from data.



Corridor-Calibrated Physics

Computes speeds and queues using historical 60th-percentile free-flow data per individual arterial, avoiding flawed city-wide averages.

Gridlock 2.0

From Blind Reaction to
Transparent Command

Thank You

Team: The Query Squad

Project: Flipkart Gridlock 2.0 Hackathon | Event-Driven Congestion Mitigation

Platform Status: Evaluated on 8,173 Events. Production Ready.